Variable Selection in Gene Regulatory Network Models

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1 Model

Gene Regulatory Network dynamics are modelled by S coupled ODEs, $\mathbf{x}'(t) = \mathbf{f}(\mathbf{x}, \theta, t)$, integrated over T time points. State variables, $\mathbf{x}(t)$, represent mRNA concentrations and ODE parameters, θ, represent biological kinetic constants. The model attempts to capture a functional relationship between a set of noisy mRNA time series observations, $\mathbf{y}(t)$, and their derivatives. The overall system is summarized by the $T \times S$ matrices: $Y_{T\times S} = X + (\epsilon_{ij})$ with independent Gaussian noise for each state $\epsilon_{ij} \sim \mathcal{N}(0, \sigma_j)$

The *parameter space*, Θ, of a model, with K parameters, $\theta_1, ..., \theta_K$, is the set of all possible values for each parameter. A particular instance of parameters, $\theta = [\theta_1, ..., \theta_K]$, is a vector in $\Theta \subset \mathbb{R}^K$. It's useful to think of a model as a map, $M : \Theta \to \mathcal{M}$, taking points/subsets of Θ to output values or points on the *model image*, $\mathcal{M}$. $\mathcal{M}$ is a K-dimensional surface embedded in $\mathbb{R}^d$, where d is the dimension of $M(\theta)$. For example, if our model is a system of ODEs with S state variables integrated over T time points then $d = S \cdot T$. A point $M(\theta)$ can be realized by concatenating the solution vectors of all state variables to create a vector of length $S \cdot T$. We will consider the following gene regulatory network model throughout:

$$
\begin{aligned}
A' &= \alpha_1 \frac{K_1^{n_1}}{K_1^{n_1} + D^{n_1}} - \beta_1 A + \gamma_1 \\
B' &= \alpha_2 \frac{A^{n_2}}{K_2^{n_2} + A^{n_2}} - \beta_2 B + \gamma_2 \\
C' &= \alpha_3 \frac{A^{n_3}}{K_3^{n_3} + A^{n_3}} + \alpha_4 \frac{B^{n_4}}{K_4^{n_4} + B^{n_4}} - \beta_3 C + \gamma_3 \\
D' &= \alpha_5 \frac{B^{n_5}}{K_5^{n_5} + B^{n_5}} + \alpha_6 \frac{C^{n_6}}{K_6^{n_6} + C^{n_6}} - \beta_4 D + \gamma_4 \\
E' &= \alpha_7 \frac{A^{n_7}}{K_7^{n_7} + A^{n_7}} - \beta_5 E + \gamma_5
\end{aligned}
\tag{1}
$$

2 Sampling Parameters

Biological models of gene regulatory processes typically have many parameters that are poorly known, either from uncertainty in data or from intrinsic unidentifiability. Intrinsic unidentifiability results from the nonlinearities of M, where certain parameter combinations, like ratios $k_1/(k_2 + X)$, with $0 \leq X \ll 1$, or combinations of degradation and basal rates $-\beta X + \gamma$, can meaninglessly wander off to infinity or zero. Parameter space is often very high dimensional (hundreds to thousands of parameters) and highly correlated. Just sampling the full posterior of 31 parameters in 1 is very challenging. A number of approaches have been developed to address these challenges in the context of systems biology models. In

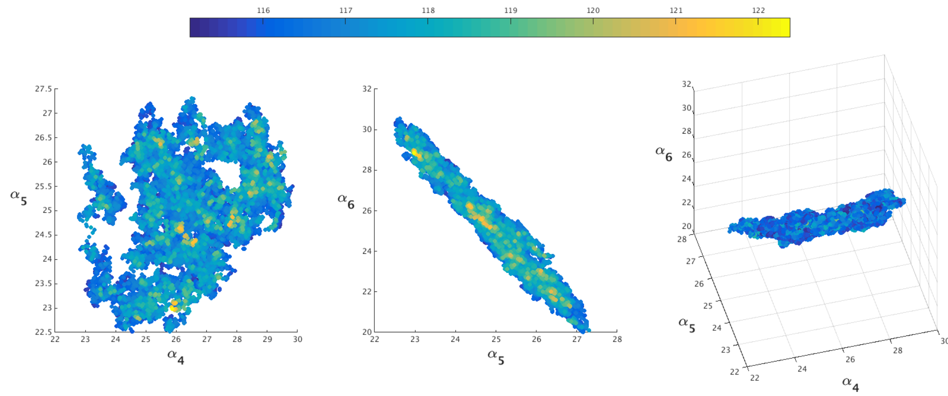

Figure 1: Correlated parameter samples ($\alpha_4 - \alpha_6$) from gene regulatory network model 1
Points colored from blue to yellow high (blue) to low (yellow) density

[2] [3] the local geometry of parameter space is exploited to deal with high dimensionality and correlations. Priors are used to control intrinsic unidentifiability. We use the 'simplified MMALA' MCMC algorithm described in [3] to collect posterior samples.

2.1 Posterior Samples

Now consider parameter samples, $\mathcal{S}$, from the posterior $\pi(\theta) \propto \widetilde{\pi}(\theta) = \exp(-C(\theta, d^\star))\, p(\theta)$ where

$$-\mathcal{L}(\theta) = -\log(\tilde{\pi}) = C(\theta, d^\star) - \log(p(\theta)) = \frac{1}{2}(M(\theta) - d^\star)^T\, \Sigma^{-1}\, (M(\theta) - d^\star) - \log(p(\theta)) \tag{2}$$

is a sum of squared residuals plus a prior term $p(\theta)$ that keeps parameters nonnegative, $\Sigma_{d\times d}$ is covariance matrix ($\Sigma = diag([(\sigma_1, ..., \sigma_1), ..., (\sigma_N, ..., \sigma_N)])$) and $d^\star \in \mathbb{R}^d$ is noisy experimental data.

2.2 Simplified MMALA

MCMC is performed using the Metropolis-Hastings algorithm with proposals generated by a discretized Langevin diffusion informed by a parameter dependent hessian of $\mathcal{L}(\theta)$, $G(\theta)$, which accounts for parameter correlations [3]. After experimentation this is the simplest MCMC scheme I found capable of sampling the posterior π efficiently

The following SDE has π as its stationary distribution:

$$d\theta(t) = \frac{1}{2}\nabla_\theta\, \mathcal{L}(\theta)\, dt + d\mathbf{b}(t) \tag{3}$$

A first order discretization of (3) and introducing a parameter dependent Hessian yields:

$$\theta^* = \theta^n + \frac{\epsilon^2}{2}G^{-1}(\theta^n)\nabla_\theta\, \mathcal{L}(\theta^n) + \epsilon\sqrt{G^{-1}(\theta^n)}z^n \tag{4}$$

where $z \sim \mathcal{N}(0, I_S)$ and $\epsilon \in \mathbb{R}$ is a step size. Each proposed parameter $\theta^\star$ of (4) has density:

$$q(\theta^*|\theta^n) = \mathcal{N}(\theta^* \,|\, \mu(\theta^n, \epsilon), \epsilon\, G(\theta^n)), \text{ where } \mu(\theta^n, \epsilon) = \theta^n + \frac{\epsilon^2}{2}G^{-1}(\theta^n)\nabla_\theta\, \mathcal{L}(\theta^n) \tag{5}$$

$G = J^T J$ where $J \in \mathbb{R}^{d\times K}$ is the Jacobian of $\mathcal{L}(\theta)$ which must be calculated using $automatic\, differentiation$ [6] [7] in the context of the posterior described in 2.1.

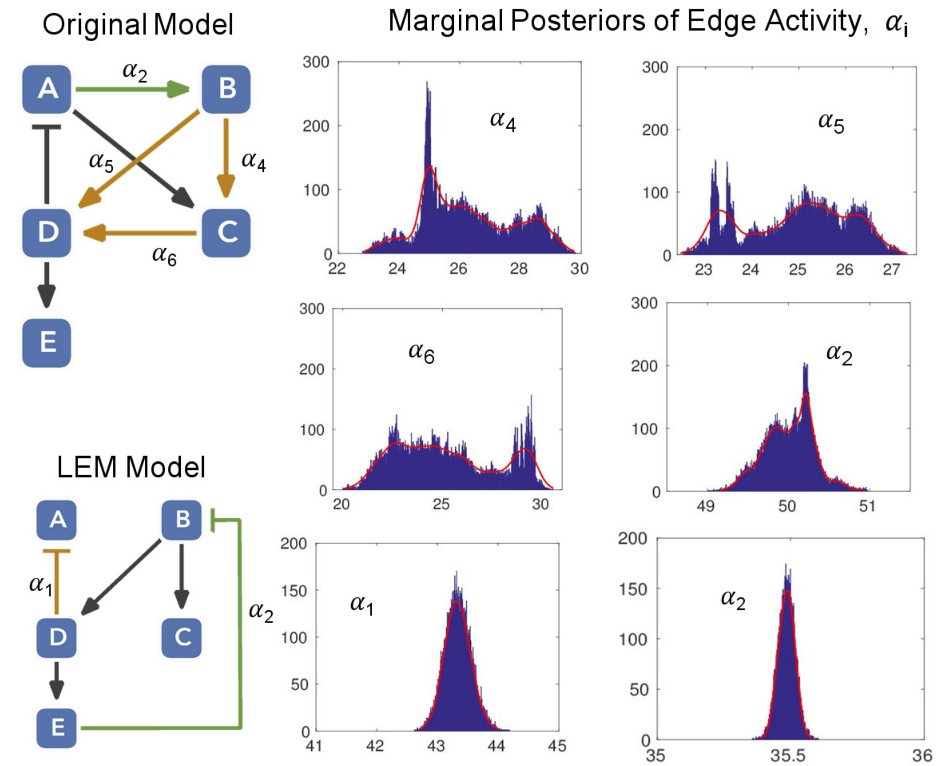

Figure 2: ($\alpha_4 - \alpha_6$) and (α_2) in the complex (top) model have the most variation and least variation respectively. α_1 and α_2 have the most and least variation in the LEM model respectively. Clearly the complex model is significantly more robust and likely preferred in a biological context.

2.3 Quantifying Robustness with Marginal Posteriors

Insights into a network's robustness can be obtained by examining marginal distributions. Under the assumption that a more robust network is more biologically fit, we can compare candidate networks based on the variance of corresponding marginal distributions [11, 12]. In 2.3 the local edge machine cannot differentiate between the two networks (top network is 1) as both models can be optimized, pointwise, to match the data equally well. MCMC sampling clearly separates the two models in 2.3 as the edge with the least variation (α_2) in the complex model has more variation than the edge with the most variation (α_1) in the sparser fragile model.

3 Variable Selection

3.1 Reversable-Jump MCMC

In addition to uncertainty in parameters, there is frequently uncertainty in network structure. This was partially addressed in section 2.3 by looking at marginal distributions. A more precise way to select between different models is to sample different structural changes in addition to parameters. As in 2.3, assume we have two candidate models, M_1 and M_2 with parameter spaces $\Theta_1 \subset \mathbb{R}^{K_1}$ and $\Theta_2 \subset \mathbb{R}^{K_2}$ respectively. The trick is to propose jumps between models in addition to parameter proposals within Θ_1 and Θ_2 [8]. Model M_i is selected over M_j if the number of iterations spent in Θ_i is greater than the number of iterations spent in Θ_j, namely if $(|\{\theta \in \mathcal{S} : \theta \in \Theta_i\}| \;/\; |\{\theta \in \mathcal{S} : \theta \in \Theta_j\}|) > 1$. There is no straightforward way to construct the proposal distribution that jumps between models, but they are 2 rules that must be followed to ensure convergence to the target distribution [8]:

1. **Dimension Matching:** Consider two models M_1 and M_2 with $\Theta_1 \subset \mathbb{R}^2$ and $\Theta_2 \subset \mathbb{R}$. For example, if a proposal from M_1 to M_2 is accepted transform $\theta^1 = (\theta_1, \theta_2) \in \Theta_1$ to $\theta^2 = (0, \theta) \in \Theta_2$ via the map $f_{12} : (\theta_1, \theta_2) \to (0, (\theta_1 + \theta_2)\ /\ 2)$. To transform $(0, \theta) \in \Theta_2$ back to $(\theta_1, \theta_2) \in \Theta_1$ we need to introduce some extra randomly generated parameters, $u^{(i)}$, 'to match dimensions' and make f_{12} bijective; we require $K_1 + r_1 = K_2 + r_2$ in the forward and reverse maps: $f_{12} : (\theta^{(1)}, u^{(1)}) \to (\theta^{(2)}, u^{(2)})$, $f_{21} : (\theta^{(2)}, u^{(2)}) \to (\theta^{(1)}, u^{(1)})$ with $u^{(1)} \in \mathbb{R}^{r_1}$ and $u^{(2)} \in \mathbb{R}^{r_2}$. For example, we can construct the reverse map as follows: $f_{21} : (0,\ \theta) \to (\theta + u,\ \theta - u)$ with u sampled from some arbitrary distribution.

2. **Volume Preservation:** The acceptance probability $\alpha(\,(M_1, \theta^1),\, (M_2, \theta^2))$ of a transition from M_1 to M_2 is calculated as:

$$\min\left(1,\ \frac{\pi(\,(M_2,\ \theta^2))\cdot j((M_2,\ \theta^2))\cdot q_2(u^2)}{\pi(\,(M_1, \theta^1))\cdot j((M_1,\ \theta^1))\cdot q_1(u^1)}\cdot\left|\frac{\partial((M_2, \theta^2))}{\partial((M_1, \theta^1))}\right|\right)$$

   π is the unormalized posterior, j is the distribution that proposes jumps between models, q_i is the density u^i is sampled from and, in particular, $\frac{\partial((M_i, \theta^i))}{\partial((M_j, \theta^j))}$ is the Jacobian of the transformation f_{ji} and its determinant corrects for the change in volume as parameters are transformed between models.

In the context of our model we create $\alpha(\,(M_1, \theta^1),\, (M_2, \theta^2))$ constrained by these 2 rules as follows:

1. **Models:** M_1 is taken to be the full system of ODEs in 1 with $\Theta_1 \subset \mathbb{R}^{31}$ and M_2 is this same system but restricted to $\Theta_2 \subset \mathbb{R}^{28}$ by setting α_6 (K_6 and n_6 are also lost), the parameter with the most variability (see 2.3), to zero.

2. **Parameter Transformations:**

   (a) $\alpha^\star = f_{12}\{([\alpha_5,\, \alpha_6],\, u^1_{\alpha^\star})\} = (\alpha_5 + \alpha_6)\cdot(1 + \ell\, u^1_{\alpha^\star})$
   (b) $K^\star = f_{12}\{(K_5,\, u^1_{K^\star})\} = K_5 \cdot (1 + \ell\, u^1_{K^\star})$
   (c) $n^\star = f_{12}\{(n_5,\, u^1_{n^\star})\} = n_5 \cdot (1 + \ell\, u^1_{n^\star})$
   (d) $\alpha_5 = f_{21}\{(\alpha^\star,\, u^2_{\alpha_5})\} = \alpha^\star \cdot (\frac{1}{2} + \ell\, u^2_{\alpha_5})$
   (e) $\alpha_6 = f_{21}\{(\alpha^\star,\, u^2_{\alpha_6})\} = \alpha^\star \cdot (\frac{1}{2} + \ell\, u^2_{\alpha_6})$
   (f) $K_5 = f_{21}\{(K^\star,\, u^2_{K_5})\} = K^\star \cdot (1 + \ell\, u^2_{K_5})$
   (g) $K_6 = f_{21}\{(K^\star,\, u^2_{K_6})\} = K^\star \cdot (1 + \ell\, u^2_{K_6})$
   (h) $n_5 = f_{21}\{(n^\star,\, u^2_{n_5})\} = n^\star \cdot (1 + \ell\, u^2_{n_5})$
   (i) $n_6 = f_{21}\{(n^\star,\, u^2_{n_6})\} = n^\star \cdot (1 + \ell\, u^2_{n_6})$

   f_{12} transforms D' (equation 1) into:

$$D' = \alpha^\star \frac{B^{n^\star}}{(K^\star)^{n^\star} + B^{n^\star}} - \beta_4\, D + \gamma_4 \tag{6}$$

   Where $u^i_{f_{ij}\{(\theta^i, u^i)\}} \sim \mathcal{U}[-1,\, 1]$ and the fraction ℓ is set to control the amount parameters are perturbed when transformed. When the parameters are optimized to the data $(\alpha_5,\, K_5,\, n_5) = (\alpha_6,\, K_6,\, n_6)$ so the parameter transformation $\alpha^\star = (\alpha_5 + \alpha_6)$ should compensate for taking α_6 out.

3. **Dimension Matching:** $K_1 + r_1 = K_2 + r_2$ as $K_1 = 31$, $r_1 = 3$, $K_2 = 28$ and $r_2 = 6$.

4. **Volume Preservation:** The Jacobians, $\mathcal{J}_{12}$ and $\mathcal{J}_{21}$ of transformations f_{12} and f_{21} follow directly from the (reordered) parameter transformations above:
$\mathcal{J}_{12} = \frac{\partial(\theta^2, u^2)}{\partial(\theta^1, u^1)} = \mathrm{diag}[\mathcal{J}_{\alpha^\star}, \mathcal{J}_{K^\star}, \mathcal{J}_{n^\star}]$ and $\mathcal{J}_{21} = \frac{\partial(\theta^1, u^1)}{\partial(\theta^2, u^2)} = \mathrm{diag}[\mathcal{J}_{\alpha_5\,\alpha_6}, \mathcal{J}_{K_5\,K_6}, \mathcal{J}_{n_5\,n_6}]$ where:

$$\mathcal{J}_{\alpha^\star}, \mathcal{J}_{K^\star} = \begin{pmatrix} 1/\ell\,\alpha^\star & 0 & 0 \\ 1 + \ell\, u_{\alpha^\star} & \ell \cdot (\alpha_5 + \alpha_6) & 1 + \ell u_{\alpha^\star} \\ 0 & 0 & 1/\ell\,\alpha^\star \end{pmatrix}, \begin{pmatrix} 1/\ell\, K^\star & 0 & 0 \\ 1 + \ell\, u_{K^\star} & \ell K_5 & 1 + \ell\, u_{K^\star} \\ 0 & 0 & 1/\ell\, K^\star \end{pmatrix}$$

$$\mathcal{J}_{\alpha_5\,\alpha_6}, \mathcal{J}_{K_5\,K_6} = \begin{pmatrix} \ell\,\alpha^\star & \frac{1}{2} + \ell u_{\alpha_5} & 0 \\ 0 & \frac{1}{\ell\cdot(\alpha_5+\alpha_6)} & 0 \\ 0 & \frac{1}{2} + \ell u_{\alpha_6} & \ell\,\alpha^\star \end{pmatrix}, \begin{pmatrix} \ell\, K^\star & 1 + \ell\, u_{K_5} & 0 \\ 0 & 1/\ell K_5 & 0 \\ 0 & 1 + \ell\, u_{K_6} & \ell\, K^\star \end{pmatrix}$$

are ordered so the Jacobians are tridiagonal:

$$\frac{\partial(\theta^1,\, u^1)}{\partial(\theta^2,\, u^2)} = \frac{\partial(\alpha_5,\, u_{\alpha^\star},\, \alpha_6,\, \mid K_5,\, u_{K^\star},\, K_6,\, \mid n_5,\, u_{n^\star},\, n_6)}{\partial(u_{\alpha_5},\, \alpha^\star,\, u_{\alpha_6},\, \mid u_{K_5},\, K^\star,\, u_{K_6},\, \mid u_{n_5}, n^\star, u_{n_6})}$$

for explicit calculation of eigenvalues [1] and the form of $\mathcal{J}_{n^\star}$ and $\mathcal{J}_{n_5\,n_6}$ is exactly the same as $\mathcal{J}_{K^\star}$ and $\mathcal{J}_{K_5\,K_6}$ respectively. Parameters in $\frac{\partial(\theta^2, u^2)}{\partial(\theta^1, u^1)}$ have the same ordering as $\frac{\partial(\theta^1, u^1)}{\partial(\theta^2, u^2)}$.

5. **Proposing jumps:** A jump from M_i to M_k, $j_{ik} = j((M_k, \theta^k))$, is proposed with 50% probability independently of the current parameters every step of the Markov Chain.

6. **Point Mass Priors:** The balance between the benefits of edge redundancy (biologically favourable robustness) and sparsity (in selecting an interpretable model) is very difficult to reason about. Introducing point mass priors of the form:

$$p(\theta_i) = r \cdot \delta(\theta_i) + (1 - r) \cdot C_i(\psi_i) \tag{7}$$

on edges candidate for removal can help in reasoning about this trade off for different values of r. The value of r is informed by prior beliefs and data. $\delta(\theta_i)$ is a point mass that takes value 1 when $\theta_i = 0$ (zero elsewhere) and $C_i(\psi_i)$ is any parametric distribution parametrized by ψ_i. It's very difficult to generate proposals that mix well between models. Setting a large value of r helps to improve the mixing between models if π suffers a significant reduction in cost when moving to a dimensionally reduced model. In this example a single point mass prior, $p(\alpha_6) = r \cdot \delta(\alpha_6) + (1 - r) \cdot \mathcal{U}[15, 35]$ is used.

First note that despite this implementation of reversable-jump MCMC being simplified to the degree of impracticality (two models, one structural change, one change in dimension), it is still quite involved to construct correctly. Furthermore, despite this construction being correct, it does not work well in practice. The difficulty of generating good jump proposals for reversible-jump MCMC is a well documented problem and a primary reason it's avoided [9] [8]. Clearly the proposal is bound to the fine level details of the model. To get this construction to work, $-\mathcal{L}(\theta)$ had to be simplified from a sum of squared residuals 2.1 to a cost that just accounts for a change in period. The main benefit of reversible jump is that it allows the use of efficient derivative informed HMC sampling (efficient MCMC methods in general) within each model. The main, and fatal, drawback is the difficulty in choosing a proposal distribution, even just for 2 models.

[1] By solving the *continuant* recurrence relation

3.2 Independence Sampler with Point Mass Priors

If the class of candidate models under consideration for sampling is restricted to the 2^K potential ways of assigning parameters in $[\theta_1, ..., \theta_K]$, to zero, a sampling method that circumvents creating jump proposals between models is preferable. Rather than proposing jumps between models, we can adaptively tune the proposal distribution to match the target distribution. In this approach the proposal distribution is a weighted sum of some subset of 2^K possible models considered for inclusion. The form of this adaptively tuned proposal distribution at the nth adaptation follows as:

$$q_n(\theta) = \lambda\, q_0(\theta; \tilde{\psi}) + (1-\lambda) \sum_{m=1}^{M} w_{m,n} q_{m,n}(\theta; \psi_{m,n}) \tag{8}$$

The first term $\lambda\, q_0(\theta; \tilde{\psi})$, $\lambda \in (0, 1]$, is necessary for convergence[2] to the target distribution [10]. $q_0(\theta; \tilde{\psi})$ is set to $\mathcal{N}(\tilde{\mu}, \tilde{\Sigma})$ with $\tilde{\Sigma} = J(\tilde{\mu})^T J(\tilde{\mu})$ as in 2.1 at the optimal parameters $\tilde{\mu}$ =arg $\max_\theta \mathcal{L}(\theta)$ of the full model. The mixture term: $\sum_{m=1}^{M} w_m q_m(\theta; \psi_m)$ is a weighted sum, $\sum_{m=1}^{M} w_m = 1$, of the models under consideration. Each model proposal, $q_m(\theta; \psi_m)$ is of the form $\mathcal{N}(\mu_m, \Sigma_m)$ for the included parameters and point masses at zero for the excluded parameters. More specifically, consider a model represented by the binary sequence 101. The associated mixture component is the product of distributions $\mathcal{N}_1(\mu_{101}, \Sigma_{101}) \times \delta(\theta_2) \times \mathcal{N}_2(\mu_{101}, \Sigma_{101})$ where $\mathcal{N}_i(\mu_{101}, \Sigma_{101})$ is the ith marginal of a two dimensional Gaussian. The mixture component means, μ_{M_i} and variances Σ_{M_i} are initialized by maximizing $\mathcal{L}(\theta)$ with the excluded parameters set to zero: $\mu^0_{M_i}$ =arg $\max_{\theta \mathcal{L}(\theta)}$ and $\Sigma^0_{M_i} = J^T(\mu^0_{M_i})J(\mu^0_{M_i})$.

Note the proposal defined in 8 is independent of the current state of the parameters. The reason for using this independence sampler instead of a Markov chain is that the Markov chain may get stuck after visiting regions of parameter space where parameters are zeroed unless the zeroed parameters have a very small region of support near zero. Reversable-jump solves this issue by 'jumping' out of these regions. Proposals are accepted/rejected for a proposed parameter $\theta^\star$ using the standard metropolis-hastings ratio: $\pi(\theta^\star) \cdot q_n(\theta) / \pi(\theta) \cdot q_n(\theta^\star)$. The proposal $q_n(\theta)$ is adapted after every L accept/reject steps by applying the following update rules [10] to each parameter in the $(n+1)$th batch, $\theta^{1:L}_{n+1}$, sequentially:

$$\begin{aligned}
w^\ell_{m,n+1} &= w_{m,n} + s_{n+1}(W_m(\theta^\ell_{n+1}) - \overline{W}^\ell_{n+1}) \\
\mu^\ell_{m,n+1} &= \mu_{m,n} + \kappa^\ell_{m,n+1}(\theta^\ell_{n+1} - \mu_{m,n}) \\
\Sigma^\ell_{m,n+1} &= \Sigma_{m,n} + \kappa^\ell_{m,n+1}[(\theta^\ell_{n+1} - \mu_{m,n})(\theta^\ell_{n+1} - \mu_{m,n})^T - \Sigma_{m,n}]
\end{aligned} \tag{9}$$

where $\ell \in [1, .., L]$ is the index of some parameter set, $\theta^{1:L}_{n+1}$, in the $(n+1)$th batch and $\ell = L$ corresponds to the final updates, $\psi^{\ell=L}_{m,n+1} = \psi_{m,n+1}$, where $\psi_{m,n+1} = (w_{m,n+1},\, \mu_{m,n+1},\, \Sigma_{m,n+1})$. s_n is a step size sequence chosen so that $s_n \to 0$ which means adaptive scheme satisfies *diminishing adaptation*. Diminishing adaptation along with including the unadapted term $\lambda\, q_0(\theta; \tilde{\psi})$, for $\lambda > 0$, to satisfy *bounded convergence* guarantees convergence to the target distribution [10]. In 9

$$\begin{aligned}
W_i(\theta^\ell_{n+1}) &= \frac{\mathcal{N}(\theta^\ell_{n+1} | \mu_{i,n}, \Sigma_{i,n})}{\sum_{m=1}^{M} w_{m,n}\, \mathcal{N}((\theta^\ell_{n+1} | \mu_{m,n}, \Sigma_{m,n})} \\
\overline{W}^\ell_{n+1} &= \frac{1}{M} \sum_{i=1}^{M} W_i(\theta^\ell_{n+1}) \\
\kappa^\ell_{m,n+1} &= s_{n+1} w_{m,n} W_i(\theta^\ell_{n+1})
\end{aligned} \tag{10}$$

[2] to satisfy *bounded convergence*

These updates follow from minimizing the Kullback-Leiber divergence

$$\mathcal{D}_{KL}\big[\tilde{\pi}(\theta)\;||\;q(\theta\,;\,\psi)\big] = E_{\tilde{\pi}}\big[\log\frac{\tilde{\pi}(\theta)}{q(\theta\,;\,\psi)}\big] \tag{11}$$

to the unormalized target $\tilde{\pi}$ [10]. To minimize this quantity we can differentiate wrt. ψ, set it equal to zero and solve for the optimal $\psi^{\star}$. Since calculating the expectation in 11 is generally intractable, we can approximate it with Monte Carlo integration:

$$h(\psi) = \frac{\partial}{\partial\psi}E_{\tilde{\pi}}\big[\log\frac{\tilde{\pi}(\theta)}{q(\theta\,;\,\psi)}\big] \approx \frac{1}{N}\sum_{i=1}^{N} f(\theta^{i},\,\psi) \tag{12}$$

where $f(\theta^{i},\,\psi) = \frac{\partial}{\partial\psi}\big[\log\frac{\tilde{\pi}(\theta)}{q(\theta\,;\,\psi)}\big]$, and $\theta^{i} \sim \pi(\theta)$. But since θ^{i} are only approximate samples from π we can approximate $h(\psi)$ with the noisy observation $\hat{h}(\theta^{1:N};\,\psi) = h(\psi) + \xi$ for some noise ξ [10]. We can find the roots of $h(\psi) = 0$ by using its noisy approximation $\hat{h}(\theta^{1:N};\,\psi)$ with the following stochastic approximation where n indexes the sample batch, $\theta_{n}^{1:L}$, as in 9:

$$\psi_{n+1} = \psi_{n} + s_{n+1}\hat{h}(\theta_{n}^{1:L};\,\psi_{n}) \tag{13}$$

For the case of a mixture of normals, the updates follow exactly as in 9. It can be shown that 11 is convex and converges to a unique root when $q(\theta)$ is from an exponential family [10]. Given that all candidate models need to be explicitly written out, this approach will face a combinatorial explosion for a large amount of candidate models.

References

[1] Building a Network From Scratch. Francis Motta, Harer/Haase Lab.

[2] Daniels, Bryan C., et al. "Sloppiness, robustness, and evolvability in systems biology." Current opinion in biotechnology 19.4 (2008): 389-395.

[3] Mark Girolami, Ben Calderhead. Riemann manifold Langevin and Hamiltonian Monte Carlo methods. Journal of the Royal Statistical Society: Series B (Statistical Methodology) Volume 73, Issue 2, pages 123214, March 2011. doi:10.1111/j.1467-9868.2010.00765.x

[4] Transtrum M.K., Chen Y.J., Machta B.B., Gutenkunst T., and Sethna J.P. Contribution to discussion on "Riemann manifold Langevin and Hamiltonian Monte Carlo Methods". Journal of the Royal Statistical Society: B. 72 2, 123-214, (2011) (Discussion on page 199.)

[5] Calderhead, Ben, and Mark Girolami. "Estimating Bayes factors via thermodynamic integration and population MCMC." Computational Statistics Data Analysis 53.12 (2009): 4028-4045.

[6] R. Serban and A. C. Hindmarsh, "CVODES: the Sensitivity-Enabled ODE Solver in SUNDIALS," Proceedings of IDETC/CIE 2005, Sept. 2005, Long Beach, CA. Also available as LLNL technical report UCRL-JP-200039.

[7] Automatic Differentiation of Stiff ODEs. Colbert Sesanker. 2015.

[8] Green, Peter J. "Reversible jump Markov chain Monte Carlo computation and Bayesian model determination." Biometrika 82.4 (1995): 711-732.

[9] Brooks, Stephen P., Paulo Giudici, and Gareth O. Roberts. "Efficient construction of reversible jump Markov chain Monte Carlo proposal distributions." Journal of the Royal Statistical Society: Series B (Statistical Methodology) 65.1 (2003): 3-39.

[10] Ji, Chunlin, and Scott C. Schmidler. "Adaptive markov chain Monte Carlo for Bayesian variable selection." Journal of Computational and Graphical Statistics 22.3 (2013): 708-728.

[11] Kitano, Hiroaki. "Towards a theory of biological robustness." Molecular systems biology 3.1 (2007).

[12] Kitano, Hiroaki. "The theory of biological robustness and its implication in cancer." Systems Biology. Springer Berlin Heidelberg, 2007. 69-88.

[13] Transtrum M.K., Machta B.B., and Sethna J.P. The geometry of nonlinear least squares with applications to sloppy models and optimization. Physical Review E. 83 036701, (2011)

[14] Aldana, Maximino, et al. "Robustness and evolvability in genetic regulatory networks." Journal of theoretical biology 245.3 (2007): 433-448.

[15] Ranganathan, Ananth. "The levenberg-marquardt algorithm." Tutoral on LM Algorithm (2004).

[16] Wagner, Andreas. "Robustness and evolvability: a paradox resolved." Proceedings of the Royal Society B: Biological Sciences 275.1630 (2008): 91-100.